Your chunks are ready. We need to convert them into embeddings. Embeddings are representations of information like text or images. Think of them as a type of machine learning feature. This format allows algorithms to perform a semantic search when ingesting our content into the Silver layer later.

We use a model to convert our chunks into embeddings. Databricks Model Serving has set of **Foundation Model APIs** (**FMAPI**), which allow you to access and query state-of-the-art open models from a serving endpoint, see *Figure 3.26*.

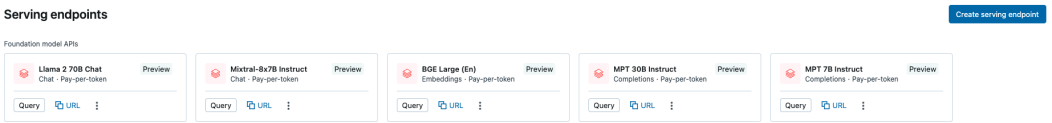


Figure 3.26 – Available models for access via FMAPI of Databricks

With Foundation Model APIs, you can quickly and easily build applications that leverage a high-quality generative AI model without maintaining your own model deployment (for more information, see *Deploy provisioned throughput Foundation Model APIs* in *Further Reading*).

The Foundation Model APIs are available in two access modes:

- **Pay-per-token**: This is the easiest way to start accessing Foundation Models on Databricks and is recommended for beginning your journey with Foundation Models.
- **Provisioned throughput**: This mode is recommended for workloads that require performance guarantees, fine-tuned models, or have additional security requirements.

Next, we leverage the DI Platform's pay-per-token capability from FMAPI that provides you with access to the `BGE_large_En` endpoint through the new functionality recently added to the `mlflow >=2.9` - mlflow deployments (previously known as AI Gateway). This functionality unifies the model serving endpoint management on Databricks.

> **Note**
>
> At the time of writing, the FMAPI is available only to US regions. If your workspace is not yet in a supported region, you can use any model of your choice (OpenAI, BERT, Llama tokenizer, etc.) to convert your content into embeddings. You may also need to fine-tune your model embedding to learn from your own context for better retrieval results.